

• elektrostatika: sile med mirujočimi el. naboji

$$\vec{F}_e = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \frac{\vec{r}}{r}$$

Coulombov zakon

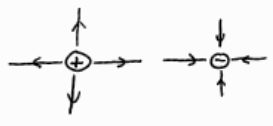
• jakost el. polja - posrednik el. sile

$$\vec{F} = e\vec{E}$$

• smer: določena s smerjo sile na pozitiven točkast naboj

• el. silnice:

- kažejo v smeri $\vec{E}$
- se ne sekajo
- niso zaključene (sklenjene)
- začnejo se v $+$, končajo v $-$ nabojem
- gostota silnic ustreza jakosti polja

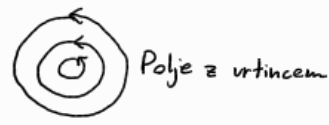


$$\oint \vec{E} \cdot d\vec{r}$$

El. cirkulacija

• za vsa statična $\vec{E}$ velja $\oint \vec{E} \cdot d\vec{r} = 0 \Rightarrow \oint \vec{E} \cdot d\vec{r} = \oint (\nabla \times \vec{E}) \cdot d\vec{S} \Rightarrow \nabla \times \vec{E} = 0$

→ statično el. polje je brez vrtincev



$$\Phi_e = \int \vec{E} \cdot d\vec{S}$$

→ če površina sklenjena $\Rightarrow$ Gaussov zakon: $\int \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$

El. pretok

$$E(\vec{r}) = -\nabla \varphi(\vec{r})$$

Električni potencial

• Princip superpozicije: $\vec{F}(\vec{r}) = \sum_i \frac{q_i q}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|^3}$ (sistem nabojev)

→ Celotna sila je vsota posameznih prispevkov sil med posameznimi nabojema.

→ Enako velja za el. polje in potencial

• Gostota naboja

$$\rho(\vec{r}) = \sum_i q_i \delta^3(\vec{r} - \vec{r}_i)$$

DISKRETNNA

lokacija naboja

$$\rho(\vec{r}) = \frac{dq}{dV}$$

ZVEZNA

$$\vec{F}(\vec{r}) = \int \rho(\vec{r}') \vec{E}(\vec{r}) d^3r'$$

$$\vec{E}(\vec{r}) = \int \frac{\rho(\vec{r}')}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} d^3r'$$

$$\varphi(\vec{r}) = \int \frac{\rho(\vec{r}')}{4\pi\epsilon_0} \frac{1}{|\vec{r} - \vec{r}'|} d^3r'$$

Posplošitev $\vec{F}, \vec{E}, \varphi$ z gostoto

• Primeri gostote naboja

1) točkast naboj

$$\rho(\vec{r}) = q \delta^3(\vec{r} - \vec{r}_0)$$

položaj naboja

2) Volumsko porazdel. naboj

$$\rho(\vec{r}) = \begin{cases} \rho_0 & \text{znotraj} \\ 0 & \text{sicer} \end{cases}$$

3) točkast dipol

$$\rho(\vec{r}) = q \delta^3(\vec{r} - \vec{r}_0) - q \delta^3(\vec{r} - \vec{r}_0 + \vec{d})$$

4) Volumsko porazdel. naboj

$$\rho(\vec{r}) = \begin{cases} \rho_0 & \text{znotraj} \\ 0 & \text{sicer} \end{cases}$$

5) Volumsko porazdel. dipoli (krogla)

$$\vec{P} = \begin{cases} \vec{P}_0 & \text{r.s.a} \\ 0 & \text{sicer} \end{cases}$$

6) Volumsko porazdel. naboj

$$\rho(\vec{r}) = \begin{cases} \rho_0 & \text{znotraj} \\ 0 & \text{sicer} \end{cases}$$

7) Volumsko porazdel. naboj

$$\rho(\vec{r}) = \begin{cases} \rho_0 & \text{znotraj} \\ 0 & \text{sicer} \end{cases}$$

8) Volumsko porazdel. naboj

$$\rho(\vec{r}) = \begin{cases} \rho_0 & \text{znotraj} \\ 0 & \text{sicer} \end{cases}$$

2) točkast dipol

$\rho(\vec{r}) = q \delta^3(\vec{r} - \vec{r}_0) - q \delta^3(\vec{r} - \vec{r}_0 + \vec{d})$

$\vec{d} \ll r - \vec{r}_0 \Rightarrow \rho(\vec{r}) \approx q \delta^3(\vec{r} - \vec{r}_0) - q \delta^3(\vec{r} - \vec{r}_0 + \vec{d})$

$\vec{P} = q\vec{d}$ dipolni moment

$\rho(\vec{r}) \approx -\vec{P} \cdot \nabla \delta^3(\vec{r} - \vec{r}_0)$

gostota dipolnega momenta (polarizacija)

* $f(\vec{r} + \Delta\vec{r}) = f(\vec{r}) + \Delta\vec{r} \cdot \nabla f(\vec{r})$

** $\nabla(\vec{P} \cdot \delta^3(\vec{r} - \vec{r}_0)) = (\nabla \cdot \vec{P}) \delta^3(\vec{r} - \vec{r}_0) + \vec{P} \cdot \nabla \delta^3(\vec{r} - \vec{r}_0)$

kar $\vec{P} = \text{konst.}$

3) Površinsko porazdeljen naboj

$\rho(\vec{r}) = \sigma(\vec{r}) \delta^2(\vec{r} - \vec{r}_0)$

površinska gost. naboja

Gaussov izrek v integralni obliki

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

(prek izreka Gauss-Ostrogradski: $\oint \vec{A} \cdot d\vec{S} = \int \nabla \cdot \vec{A} d^3r$)

Gaussov izrek v diferencialni obliki

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Poissonova enačba

$$\nabla^2 \varphi = -\frac{\rho}{\epsilon_0}$$

če $\rho(\vec{r}) = 0$: Laplaceova enačba

$$\nabla^2 \varphi = 0$$

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$$\nabla^2 \varphi = -\frac{\rho}{\epsilon_0} \rightarrow \text{rešitev išemo kot konvolucijo: } \varphi(\vec{r}) = \int \frac{1}{|\vec{r} - \vec{r}'|} \rho(\vec{r}') d^3r' \rightarrow \text{predp, da taken rešitev \text{3.}}$$

Grešno v Fourierov prostor:

$$\rightarrow \nabla^2 \left[\int \frac{d^3 k}{(2\pi)^3} e^{ik(\vec{r}-\vec{r}')} G(k) \right] = -\frac{1}{\epsilon_0} \int \frac{d^3 k}{(2\pi)^3} e^{ik(\vec{r}-\vec{r}')} = -\int \frac{d^3 k}{(2\pi)^3} \left[\nabla^2 e^{ik(\vec{r}-\vec{r}')} G(k) + \frac{1}{\epsilon_0} e^{ik(\vec{r}-\vec{r}')} \right] = 0$$

→ dobimo podobno enačbo,
le da je nova en. za točkast
maboj. Greenova funkcija
je rešitev Pois. en.
za točkast maboj v $\mathbb{R}^1$.

$$\rightarrow \int \frac{d^3 \vec{r}}{(2\pi)^3} e^{i \vec{k} \cdot (\vec{r} - \vec{r}')} \left[-\frac{1}{\epsilon_0} G(\vec{k}) + \frac{1}{\epsilon_0} \right] = 0 \rightarrow \boxed{G(\vec{k}) = \frac{1}{\epsilon_0 k^2}} \rightarrow G(\vec{r} - \vec{r}') = \frac{1}{\epsilon_0} \int \frac{d^3 \vec{k}}{(2\pi)^3} e^{i \vec{k} \cdot (\vec{r} - \vec{r}')} \frac{1}{k^2} = \frac{1}{\epsilon_0} \int \int_0^\pi \int_0^{2\pi} d\varphi \sin \theta d\theta dk e^{i \vec{k} \cdot (\vec{r} - \vec{r}') \cos \theta} \frac{1}{k^2} = \frac{1}{\epsilon_0} \int_0^\pi \int_{-1}^1 d(\cos \theta) e^{i k |\vec{r} - \vec{r}'| \cos \theta} = \frac{1}{\epsilon_0} \int_0^\pi \frac{\sin x}{x} dx = \frac{1}{4\pi \epsilon_0 |\vec{r} - \vec{r}'|}$$

$$\Rightarrow \varphi(\vec{r}) = \int \frac{\rho(\vec{r}')}{4\pi\epsilon_0 |\vec{r}-\vec{r}'|} d^3\vec{r}' \quad \begin{matrix} \text{3 pl. reš.} \\ \text{Poissonova} \\ \text{enačbe} \end{matrix} \rightarrow \vec{E} = -\nabla\varphi = -\nabla \left(\int \frac{\rho(\vec{r}')}{4\pi\epsilon_0 |\vec{r}-\vec{r}'|} d^3\vec{r}' \right) = - \int \frac{\rho(\vec{r}')}{4\pi\epsilon_0} \left(\nabla \frac{1}{|\vec{r}-\vec{r}'|} \right) d^3\vec{r}'$$

• Earnshawjev teorem: Naboj točkastih nabojev ne more nikoli biti v stabilnem ravnovesju samo kot posledica elektrostatskih interakcij

Posledica: el. potencial v praznem prostoru nima maksimumov in minimumov

Thysson problem: Elektrostatiska energijen ima lahko minimume, če nabije ogradiš na površini

→ npr. 2 naboja na sferi: $\odot$ (odlučivati se za dijelove od drugoga)

$$\rightarrow \boxed{W_e = e\psi}$$

→ Za zvezno porazd. naboj: $W = \int \rho(\vec{r}') \phi(\vec{r}') d^3\vec{r}'$

• Coltura el. energija (energija polja): $dW = \int d\varphi \cdot \alpha \varphi(\vec{r}') d^3\vec{r}' \rightarrow W = \int_{\alpha=0} dW = \boxed{\frac{\alpha}{2} \int \varphi(\vec{r}') \varphi(\vec{r}') d^3\vec{r}'}$
 $\downarrow$ $\downarrow$
 q ustunja q

Gostota el. en. polja : $[W] = \frac{1}{2} \int \rho \varphi d^3r = \frac{1}{2} \left(\epsilon_0 (\nabla \cdot \vec{E}) \varphi(r) d^3r \right) = \frac{\epsilon_0}{2} \int [\nabla \cdot (\varphi \vec{E}) - \nabla \varphi \cdot \vec{E}] d^3r = \dots$

Sila kot funkcional el. polja:

$$\vec{F} = \int \rho(\vec{r}) \vec{E} d^3r = \int (\nabla \cdot \vec{E}) \vec{E} d^3r =$$

celotno el. polje (vključuje tudi polje, ...)

za V vezano cel $R^3 \Rightarrow 1$ člen = 0

$$k_i \text{ ga ustvari } f(r_i))$$

• Napetostni tenzor

$$F_i = \oint_{\partial V} T_{ik} n_k dS = \int_V \frac{\partial T_{ik}}{\partial x_k} d^3x$$

$$T_{ik} = \epsilon_0 \left(E_i E_k - \frac{1}{2} E^2 \delta_{ik} \right)$$

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- Multipolni razvoj el. potenciala

$$\psi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r}-\vec{r}'|} d^3r' = \frac{1}{4\pi\epsilon_0} \frac{1}{r} \int \rho(\vec{r}') d^3r' + \frac{1}{4\pi\epsilon_0 r^3} \int \rho(\vec{r}') d^3r' \vec{r} \cdot \vec{r}'$$

- Multipolni razvoj elektrostatske energije

$$W = \int_V \rho(\vec{r}) \varphi(\vec{r}) d^3\vec{r} = \varphi(\vec{r}_0) \int_V \rho(\vec{r}) d^3\vec{r} + \nabla \cdot \varphi(\vec{r}_0) \int_V (\vec{r} - \vec{r}_0) \rho(\vec{r}) d^3\vec{r} = \underline{\varphi(\vec{r}_0) \cdot Q + p_0 \cdot E(\vec{r}_0)}$$

- Sila v zun. el. polju

$$\begin{aligned} dW &= \vec{F} \cdot d\vec{r} \\ e \nabla \varphi(\vec{r}_e) d\vec{r} + \nabla(\vec{p}_e \cdot \vec{E}(\vec{r}_e)) d\vec{r} &= \left[e \nabla \varphi(\vec{r}) + (\vec{p}_e \cdot \nabla) \vec{E}(\vec{r}_e) \right] d\vec{r} \\ \nabla(\vec{p}_e \cdot \vec{E}(\vec{r}_e)) &= \vec{p}_e \cdot \nabla \vec{E}(\vec{r}_e) + (\vec{p}_e \cdot \nabla) \vec{E}(\vec{r}_e) \end{aligned}$$

Navor v zunj. el. polju

$$\rightarrow \boxed{\vec{M} = \vec{p}_0 \times \vec{E}(\vec{r}_0)}$$